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The House Edge Without a House: Insurance for Gamblers

A pre-registered test of whether a fund can profit by selling longshot bets on Kalshi and Polymarket, and of what crash insurance from the same markets costs a stock portfolio.

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ABSTRACT

People who buy cheap longshot contracts on prediction markets lose money: on Kalshi, buyers of contracts priced at ten cents or less lose close to half of their stake. This paper asks whether a fund could take the other side of those bets, as a bookmaker would, and whether the same markets can insure a stock portfolio. The plan, hypotheses and tests were fixed before the data were examined. Part A uses 148,939 Kalshi longshots in 52,731 events from July 2021 to October 2026 and 23,911 Polymarket longshots from March 2023 to January 2026. A taker who buys “No” on every contract priced at one to ten cents a day before the close earns at most +0.06% per dollar on Kalshi after fees (95% interval [−0.05%, 0.17%]), an upper bound because it assumes buying at the last traded price; paying one cent more gives −0.97%. On Polymarket it loses 1.56%. The bias is real, but it is collected by makers: Kalshi makers who sold longshots earned +1.89%, 1.57 points more than takers. An exploratory split finds a taker edge only in liquid markets, and a fund trading it invested about 1% of its capital on an average day.

Part B treats crash insurance as a longshot. A one-day contract bundle on Kalshi's S&P 500 and Nasdaq-100 ranges, paying $1 if the index closes more than 1% below the market's forecast, cost 28.5 cents and paid out on 8.6% of 291 index-days: buyers lost 70% per dollar. The insurance pays when stocks fall, but it lowers a stock portfolio's Sharpe ratio and does not measurably reduce its worst days. Prediction markets reward whoever plays the house; on these platforms, a fund can be that house only at the margin.

Keywords: *prediction markets; favourite–longshot bias; Kalshi; Polymarket; market making; portfolio insurance; pre-registration*

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1. INTRODUCTION

Every casino, bookmaker and lottery operator lives on the same arithmetic: customers pay slightly more for a bet than it is worth, and the house collects the difference on millions of small wagers. The price of a long shot carries the largest mark-up. In horse racing, a horse quoted at 100 to 1 wins far less often than once in a hundred races; in sports betting and lotteries, the cheapest tickets lose the most per dollar. This pattern, the favourite–longshot bias, has been documented for more than seventy years (Griffith, 1949; Thaler & Ziemba, 1988; Snowberg & Wolfers, 2010).

Prediction markets were meant to be different. On Kalshi, a US exchange regulated by the Commodity Futures Trading Commission, and on Polymarket, a blockchain-based exchange, there is no bookmaker setting the odds: traders buy and sell contracts that pay $1 if an event happens, and the price is set by whoever is willing to trade. Yet the bias survives. Bürgi, Deng and Whelan (2026) show that on Kalshi, contracts costing ten cents or less lose most of their buyers' money, and that the traders who post prices (makers) do much better than those who accept them (takers).

If longshot buyers overpay, someone is being overpaid. The question of this paper is whether that someone could be a fund: a “house without a house” that systematically sells longshots on prediction markets, collects the premium and diversifies the occasional loss across thousands of unrelated events. The title's second half points to the mirror image. Buying a longshot is buying insurance against an unlikely event; a fund that holds stocks may want exactly that kind of insurance against a market fall. Part B therefore asks whether Kalshi's stock-index contracts can insure a stock portfolio, and at what price.

The study is pre-registered. Its hypotheses, sample rules, fee assumptions, tests and robustness checks were written down on 27 October 2026, before any data were downloaded; Part B was added on 30 October 2026, before any of its prices or returns had been computed. Every later change is listed in Appendix C with the reason and whether results had been seen. The sample covers every settled Kalshi market from July 2021 to 27 October 2026 and every resolved two-outcome Polymarket market from March 2023 to January 2026.

Four findings stand out. First, the bias is real but small when measured where a fund could act. A day before the close, a taker who buys “No” on every Kalshi longshot earns +0.06% per dollar after fees at the last traded price, a result statistically indistinguishable from zero, and an upper bound: paying the spread, even one cent, turns it into a loss (−0.97%). The

large losses documented in earlier work come mostly from the last trades before the close, when the outcome is often already known. On Polymarket, selling longshots as a taker loses 1.56% per dollar: longshots there are, if anything, underpriced.

Second, the premium goes to makers. Kalshi makers who sold longshots earned +1.89% per dollar, 1.57 percentage points more than takers trading in the same price band. Third, in an exploratory split, a taker edge appears in busy markets (+0.92% per dollar at the last price where at least 1,000 contracts traded the previous day), but it is mostly the spread, and it cannot absorb much money: a $1 million fund limited to a tenth of each market's volume invested on average 1.1% of its capital. Its losses are spread over unrelated events, so it does not have the crash-prone profile of a seller of volatility, and it moved independently of the stock market.

Fourth, crash insurance is a very expensive longshot. A bundle of S&P 500 and Nasdaq-100 range contracts paying $1 if the index closes more than 1% below the market's own forecast cost 28.5 cents and paid on 8.6% of days; its buyers lost 70% per dollar. It does pay when stocks fall, but a stock fund that buys it daily ends up with a lower Sharpe ratio and no measurable reduction in its worst days.

The rest of the paper is organised as follows. Chapter 2 reviews the literature. Chapter 3 explains how the two markets work, Chapter 4 describes the data and Chapter 5 the method. Chapters 6 and 7 report the results of Parts A and B. Chapter 8 discusses what they mean, and Chapter 9 sets out the limitations and concludes.

2. Literature Review

2.1 The favourite–longshot bias

Griffith (1949) first showed that American racegoers bet too much on long shots: horses with long odds won less often than their odds implied, while favourites won slightly more. The pattern has since been found in racetrack betting in many countries, in lotteries and in sports betting (Thaler & Ziemba, 1988; Ottaviani & Sørensen, 2008). Two explanations dominate. In the first, bettors enjoy risk and pay for the small chance of a large win. In the second, they misperceive probabilities and overweight small ones. Using US racetrack data, Snowberg and Wolfers (2010) find that misperception fits the data better. A third explanation comes from the bookmaker's side: when some bettors hold private information, a bookmaker protects itself by lowering the odds on long shots, where insiders do the most damage (Shin, 1993). Whelan (2024) shows that risk-averse bookmakers produce the same pattern in a competitive fixed-odds market.

Sports betting markets also behave like financial markets in other ways. Moskowitz (2021) finds momentum and value patterns in betting lines, but returns that are too small to survive the bookmaker's margin. That result anticipates one of this paper's themes: a pricing anomaly can be real without being exploitable once trading costs are paid.

2.2 Prediction markets

Wolfers and Zitzewitz (2004) describe prediction markets as a way to aggregate dispersed information into a price that can be read as a probability. Their record is good but not perfect. Page and Clemen (2013) find that prediction-market prices are well calibrated for events close in time but biased toward 50% for distant ones, so that long shots far from resolution are overpriced. Diercks, Katz and Wright (2026) show that Kalshi's macroeconomic markets forecast Fed decisions and inflation as well as surveys and futures, and offer full probability distributions rather than point forecasts.

The closest study is Bürgi, Deng and Whelan (2026). Using Kalshi trades from 2021 to April 2025, they find an average return of about −20% across contracts, losses of more than 60% on contracts costing ten cents or less, and much better returns for makers (−9.64% on average) than for takers (−31.46%). They conclude that the bias reflects the behaviour of takers who buy long shots. This paper starts by reproducing their main figures, then asks

whether a fund could earn the other side, out of sample and after fees, on both Kalshi and Polymarket.

2.3 Selling insurance and selling volatility

Selling long shots resembles selling insurance or out-of-the-money options: frequent small gains and occasional large losses. In option markets, sellers of index puts have historically earned a premium. Coval and Shumway (2001) find that S&P 500 puts earn large negative returns for their buyers, Bakshi and Kapadia (2003) attribute part of option prices to a negative volatility risk premium, and Carr and Wu (2009) measure that premium across assets. Bondarenko (2014) shows that put prices are too high to be explained by standard asset-pricing models. The same evidence implies that buying protection is costly: Israelov (2019) finds that protective puts reduce the risk of an equity portfolio by less than their cost reduces its return. Part B of this paper asks the same question of a new kind of protection, bought one day at a time on a prediction market.

3. How the Markets Work

3.1 Kalshi

Kalshi lists binary contracts on events: whether a team wins, whether inflation exceeds a threshold, where the S&P 500 will close. A contract pays $1 if the event happens and nothing otherwise, and trades between 1 and 99 cents. Buying "Yes" at p is the same position as selling "No" at $1 - p$. Contracts are grouped into events (all the candidates of one election, all the ranges of one day's S&P 500 close), and events into series. Trading is through a central limit order book: a *maker* posts an order that waits in the book; a *taker* accepts a waiting order and trades immediately.

Kalshi charges takers a fee that is largest for contracts near 50 cents and smallest for long shots (Kalshi, 2026):

$$F = \text{RoundUp}_{\text{cent}} [\, 0.07 \times C \times P \times (1 - P) \,] \qquad (1)$$

where P is the price in dollars and C the number of contracts: the whole product, including the 0.07, is computed first and then rounded up to the next cent ($\text{RoundUp}_{\text{cent}}$ above). A series can carry a multiplier on this formula. Makers paid no fee until 2025; since then, some series charge makers a quarter of the taker rate. Because of the rounding, the fee per contract depends on the size of the order. Bürgi et al. (2026) describe their fee as computed on an order of 100 contracts, and the plan does the same: on a 5-cent contract, $0.07 \times 100 \times 0.05 \times 0.95 = \$0.3325$, which rounds up to 34 cents per 100 contracts, or 0.34 cents each. Section 4.3 shows, however, that their published figures are matched by fees computed for single contracts.

3.2 Polymarket

Polymarket works the same way for the trader but settles on the Polygon blockchain, in a dollar-linked token. Its markets have two outcomes, often named after teams or answers rather than Yes and No, and most trading has been fee-free; fees appeared on some short-term markets in 2026. Every trade is recorded on the blockchain, with the addresses of both sides and the fee paid. The records contain a trap: each trade appears twice, once from the side of each maker order and once from the side of the taker's order, with the exchange contract recorded as the taker (Slivkoff, 2025). Counting both records doubles trading volume and

makes it impossible to tell who took whose price. This paper keeps one record per trade and uses the second record only to identify the taker.

3.3 Index range contracts

For each trading day, Kalshi lists about thirty contracts on where the S&P 500 and the Nasdaq-100 will close: a set of adjacent ranges (for example, “between 6,825 and 6,849.99”), plus one contract for a close below the lowest range and one for a close above the highest. Exactly one of them pays. Buying every range below some level gives a contract that pays $1 if the index closes below that level: a one-day digital put. Since November 2025, these contracts are usually listed the evening before, a median of 18 hours before the close; on 39% of index-days some ranges were listed earlier, which is when a price 24 hours before the close can exist. Before November 2025, most were listed two to three days ahead.

4. DATA

4.1 Sources

Kalshi data up to 23 November 2025 come from the public prediction-market dataset of Becker (2026), which contains every Kalshi market and trade. Markets that closed after that date were downloaded from Kalshi's public interface (Kalshi, 2026): the list of settled markets and, for each, its hourly price history, from which the last traded price before each entry time is taken. Every S&P 500 and Nasdaq-100 range contract from 24 November 2025 to 27 October 2026 was downloaded in the same way, with no volume filter. Polymarket data are the blockchain trade records in Becker's dataset, which end on 25 January 2026. The US stock market return and the risk-free rate come from Kenneth French's data library (Fama & French, 1993), up to 30 September 2026.

Table 1. Samples

Sample	Period	Markets	Use
Kalshi, trade data	Jul 2021 – 23 Nov 2025	554,233	Validation, Parts A and B
Kalshi, price histories	24 Nov 2025 – 27 Oct 2026	947,336	Part A
Kalshi index ranges	24 Nov 2025 – 27 Oct 2026	13,980	Part B
Polymarket	Mar 2023 – 25 Jan 2026	381,584	Part A
US market, risk-free rate	Jul 2021 – Sep 2026	–	Fund, Part B

Kalshi trade data: every settled binary market (554,233 with at least one trade). Price histories: settled binary markets with at least 1,000 contracts traded, open at least 24 hours, excluding multi-leg combination contracts. Each Kalshi market is one binary contract. Polymarket: resolved two-outcome markets, each with two outcome tokens; 235,190 of them had at least $1,000 of volume; 229,884,276 trades counted once. Source: Author's own calculations.

4.2 Sample rules

The rules follow Bürgi et al. (2026). A Kalshi market enters Part A if it settled Yes or No, was open for at least 24 hours, and traded at least 1,000 contracts. Markets in series that reset every hour or every fifteen minutes are excluded, as are multi-leg combination contracts after November 2025, which pay only if several outcomes all occur and make up three quarters of Kalshi's markets in 2026. A Polymarket market enters if it resolved with one outcome priced above 99 cents and the other below 1 cent, and had at least $1,000 of volume. Because Polymarket's scheduled end dates are often not when trading stops, its “close” is the market's last trade; the scheduled end date is used as a check.

4.3 Validation

Before any test, the data were checked against Bürgi et al. (2026). With their filters, the sample contains 45,372 contracts in 12,324 events, against their 46,282 and 12,403, and 181,894 daily price observations against their 156,986. The small differences are expected: their sample also drops markets whose final bid–ask spread exceeded 20 cents, which the trade data cannot see, and they take daily prices at the clock time of each market's last trade rather than its closing time.

Table 2. Validation against Bürgi, Deng and Whelan (2026)

	Published	This paper
Contracts / events	46,282 / 12,403	45,372 / 12,324
Average return, before fees	about −20%	−14.1%
Average return, after fees (100-contract orders)	–	−16.5%
Average return, after fees (single contracts)	–	−22.0%
Contracts at 10 cents or less, before fees	loss of more than 60%	−41.3%
Contracts at 10 cents or less, after fees (100-contract orders)	–	−45.1%
Contracts at 10 cents or less, after fees (single contracts)	–	−59.5%

Kalshi, launch to April 2025; markets open at least 24 hours with at least 1,000 contracts traded, excluding hourly and 15-minute series. Returns per contract at the last trade before the close and at the same time on each of the ten previous days, for both Yes and No contracts. Source: Author's own calculations.

The validation is half met. The average after the plan's fee is −16.5%, inside the pre-registered range of −30% to −10%, but contracts costing ten cents or less lose 45.1% rather than more than 50%. The published figures are matched almost exactly when the fee is computed for a single contract, rounded up to a whole cent (−22.0% and −59.5%), which suggests that the difference lies in how fees were applied rather than in the data. The paper's tests use the 100-contract fee fixed in the plan, which is the cheaper and more realistic assumption for a fund; with single-contract fees, every return to selling long shots would be lower.

5. METHOD

5.1 The strategy

At a fixed time before each market closes, 24 hours in the main test, the fund looks at the last traded price. If “Yes” costs between 1 and 10 cents, the fund buys “No” at 100 minus that price, as a taker, and holds the contract until it settles. On Polymarket, where outcomes are rarely called Yes and No, the long shot is whichever outcome costs 1 to 10 cents. The return on one contract is

$$R = (Y - P - F) / (P + F) \qquad (2)$$

where Y is 1 if the fund's side wins, P the price paid and F the fee per contract. Every contract counts equally, so the mean of R is the return per dollar of a fund that puts the same amount in each. Buying at 100 minus the last traded Yes price assumes the fund pays no spread; Section 6.3 measures what the bid and ask imply.

The maker version uses no assumption about fills. Every recorded trade in which a taker bought a long shot at 1 to 10 cents is a trade in which a maker sold it, that is, bought the other side. The maker's return comes from those trades, weighted by contracts, with maker fees where they applied. The taker comparison uses the trades in which the taker bought the favourite side at the same prices.

5.2 Hypotheses

Table 3. Hypotheses of Part A

	Hypothesis	Pass condition
V	The data reproduce Bürgi et al. (2026)	Mean return after fees between −30% and −10%; contracts at 10 cents or less lose more than 50%
H1	Buying “No” on long shots earns a positive return after taker fees (primary)	Mean return > 0, one-sided $p < 0.05$
H2	Selling long shots as a maker earns more than zero and more than as a taker	Both one-sided $p < 0.05$
H3	The fund has the profile of a volatility seller	Monthly returns negatively skewed; worst month loses more than six times the median month gains
H4	The fund diversifies a stock portfolio	Correlation with the market below 0.3 in absolute value; no loss on average in the 10% worst market months
H5	H1 holds after April 2025	H1's test on contracts closing from May 2025

H1 is tested at 5%; H2 to H5 are corrected together with the Holm (1979) procedure. Pass conditions as in the plan of 27 October 2026. After the results were seen, H3 was made stricter: it is labelled Supported *only if the skewness is also significantly negative (Appendix C, item 15; Section 6.5). Source: Author's own calculations.*

Table 4. Hypotheses of Part B

	Hypothesis	Pass condition
HB1	Crash insurance on Kalshi is expensive: buyers lose money after fees (primary)	Return per dollar < 0, one-sided p < 0.05
HB2	The insurance pays when stocks fall	Correlation of its daily profit with the market below −0.3, interval excluding 0
HB3	It lowers the portfolio's tail risk	Smaller 5% expected shortfall of daily returns, one-sided p < 0.05
HB4	It is worth its cost	Higher Sharpe ratio, one-sided p < 0.05
HB5	HB1 holds after April 2025	HB1's test on days from May 2025

HB1 is tested at 5%; HB2 to HB5 are corrected together with the Holm (1979) procedure. Source: Author's own calculations.

5.3 Inference and power

Contracts within one event are not independent: when one candidate wins, all the others lose. Confidence intervals and p-values for H1, H2 and H5 therefore come from a bootstrap that resamples whole events, 10,000 times, with a fixed seed. Polymarket's data have no event identifier, so its contracts are grouped by the day of their last trade, a coarser grouping that widens the intervals. The fund's statistics and Part B resample blocks of consecutive months or days (Politis & Romano, 1994). p-values are percentile p-values: the share of bootstrap draws on the wrong side of zero. Each hypothesis is labelled *Supported* if it meets its pass condition, *Contradicted* if its 95% interval lies entirely on the other side, and *Inconclusive* otherwise. Because the tests are one-sided at 5%, a result can be *Supported* while its two-sided 95% interval just touches zero. Results that were not planned in advance are marked as exploratory.

A “No” contract bought at 96 cents gains about 4% when it wins and loses everything when it loses, so returns are very noisy. With N independent events, the smallest mean return the test can detect with 80% power is about 2.5 standard deviations divided by $\sqrt{N}$. With 52,731 Kalshi events, that is 0.21 percentage points; with 934 Polymarket clusters, it is 1.95 points. A result smaller than these is reported as inconclusive, not as proof that no edge exists.

5.4 The fund

The fund starts with $1 million on the first day of each platform's sample. Each day it settles the contracts that closed, then divides its free cash equally among the day's eligible

contracts, subject to two limits: no more than 1% of its capital in any one event, and no more than 10% of the contracts traded in the 24 hours before entry, so that it never buys more than the market offered. Contracts are held to settlement and carried at cost; idle cash earns the risk-free rate; there is no leverage. Monthly returns in excess of the risk-free rate are compared with the US stock market.

5.5 Crash insurance (Part B)

At entry, 24 hours before the close in the main test, the fund reads the market's median forecast from the range prices: the level at which the cumulated prices, ordered from the lowest range and divided by their sum, reach one half. It then buys “Yes”, as a taker, on every range whose upper bound is at least d below that median, with d = 1%. The bundle pays $1 if the index closes in any of those ranges. The portfolio holds $1 of US stocks and h = 0.01 bundles per dollar, so that the insurance pays 1% of the portfolio on a bad day:

$$r_{*h*} = r_{*m*} + h \times (Y - C) \qquad (3)$$

where r_{*m*} is the market's return that day, Y the payout and C the bundle's cost including fees. Each index-day is a separate observation: on a date with prices for both the S&P 500 and the Nasdaq-100, the portfolio holds 0.01 bundles on each index, and each enters the tests as its own day, with the same stock return. The cost per insured day and HB1, HB2 and HB5 are therefore per index-day; HB3 and HB4 compare the expected shortfall and Sharpe ratio over the same index-days, and the bootstrap resamples dates, keeping both index-days of a date together. Only days with prices 24 hours before the close are insured. The robustness checks vary d (0.5%, 1%, 2%), h (0.005, 0.01, 0.02), the index, the fees and the entry time.

6. Results: Selling Long Shots

6.1 Do long shots win as often as their price says?

Figure 1 shows, for each price from 1 to 10 cents, the share of contracts that won, separately for the two Kalshi data periods. Before November 2025, long shots 24 hours before the close won 2.94% of the time at an average price of 3.55 cents: they were overpriced, though far less than at the last trade before the close, when many outcomes are effectively known. From November 2025 they won 3.65% at 3.90 cents, close to their price. Polymarket's long shots won slightly more often than their price said.

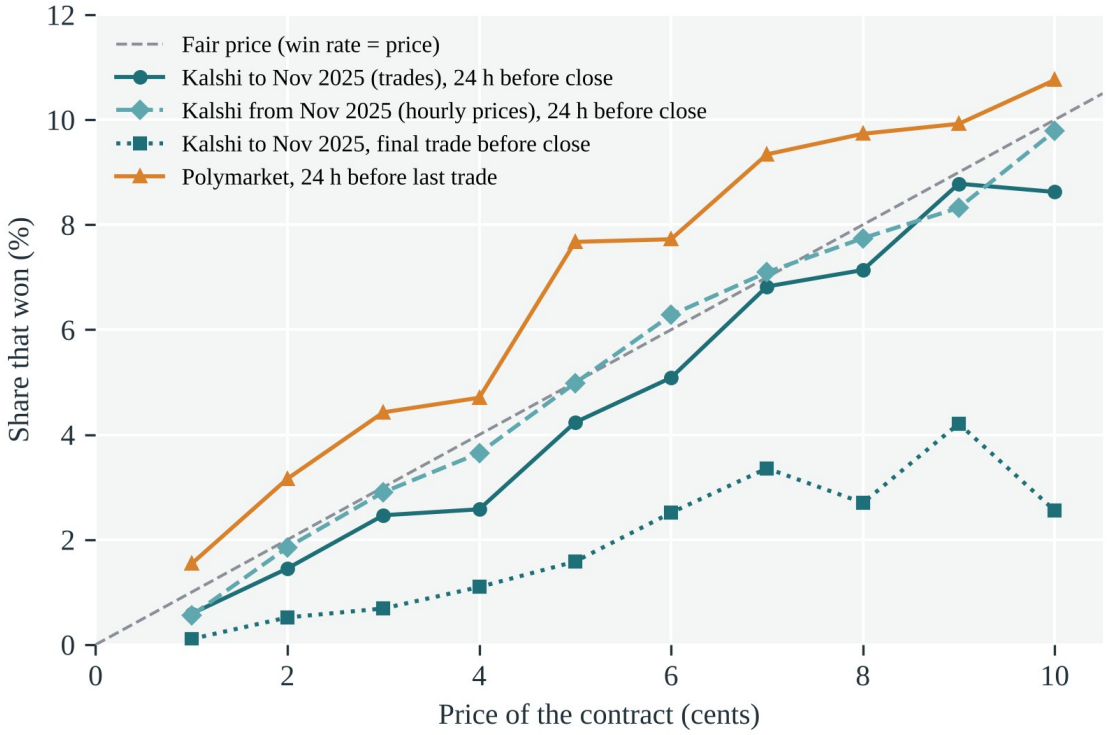


Figure 1. Share of 1–10 cent contracts that won, by price. Kalshi: markets open at least 24 hours with at least 1,000 contracts traded; before November 2025 from trade data, after from hourly price histories. Polymarket: markets with at least $1,000 of volume. Prices with at least 30 contracts. Figure produced using Matplotlib (Hunter, 2007).

Table 5. Yes buyers' return on 1–10 cent contracts by time before the close (Kalshi, to November 2025)

Time before close	Contracts	Average price	Share that won	Return before fees
7 days	7,202	2.98 c	1.99%	−43.4%
24 hours	29,723	3.55 c	2.94%	−26.9%
1 hour	60,522	2.36 c	1.37%	−58.7%
Last trade	71,774	1.85 c	0.44%	−84.2%

Trade-based sample; last traded price at or before each time. Source: Author's own calculations.

Table 5 puts numbers on this. A day before the close, Yes buyers of 1–10 cent contracts lose 26.9% before fees; at the last trade, they lose 84.2%. The largest losses are concentrated in the final hours. A loss of 27% on a 3.55-cent contract is also small in cents: for the seller, who stakes about 96 cents, it is a gain of about 0.6% before fees.

6.2 Selling long shots as a taker (H1, H5)

Table 6. Selling 1–10 cent long shots as a taker, 24 hours before the close, at the last traded price

Sample	Contracts	Clusters	Mean return	95% interval	p	Label
Kalshi, Jul 2021 – Oct 2026 (H1)	**148,939**	**52,731**	**+0.06%**	**[−0.05%, 0.17%]**	**0.143**	**Inconclusive**
to April 2025	13,051	5,397	+0.13%	[−0.19%, 0.43%]	0.212	Inconclusive
May 2025 – Oct 2026 (H5)	135,888	47,349	+0.05%	[−0.06%, 0.17%]	0.185	Inconclusive
May – Nov 2025, trade data	16,672	8,559	+0.60%	[0.33%, 0.85%]	<0.001	Exploratory
Nov 2025 – Oct 2026, hourly prices	119,216	38,795	−0.02%	[−0.15%, 0.11%]	0.637	Exploratory
without the combination contracts listed before Nov 2025	143,703	48,268	+0.05%	[−0.06%, 0.17%]	0.184	Exploratory
Polymarket, Mar 2023 – Jan 2026 (H1)	**23,911**	**934**	**−1.56%**	**[−2.01%, −1.13%]**	**1.000**	**Contradicted**
to April 2025	5,433	664	−0.92%	[−1.68%, −0.22%]	0.996	Contradicted
May 2025 – Jan 2026 (H5)	18,478	270	−1.74%	[−2.27%, −1.23%]	1.000	Contradicted
Yes/No markets only	21,490	929	−0.52%	[−0.91%, −0.15%]	0.998	Exploratory
24 hours before scheduled end	19,742	910	−1.52%	[−2.04%, −1.03%]	1.000	Exploratory

Return per dollar after fees, buying at 100 minus the last traded Yes price. Clusters are events on Kalshi and days of last trade on Polymarket. One-sided percentile p-values for a mean above zero, cluster bootstrap with 10,000 draws. Rows marked exploratory are not pre-registered tests. Source: Author's own calculations.

Table 6 reports the primary test. On Kalshi, buying “No” on every long shot 24 hours before the close earns +0.06% per dollar after fees, with a 95% interval of [−0.05%, 0.17%]. The test can detect an effect of 0.21 points, and the interval rules out anything larger than 0.17%: H1 is *Inconclusive*. The “No” side wins 96.5% of the time, just enough to cover an average price of 96.2 cents and a fee of 0.25 cents. H5 gives the same answer (+0.05%), but it covers 91% of the H1 sample and so adds little independent evidence; the contrast with the period to April 2025 is the more informative split.

The average hides a break. Before November 2025, the trade data give +0.39% per dollar (May to November 2025 alone: +0.60%); after it, the hourly price histories give −0.02%, and they make up 80% of the sample. Two things changed at that date, and both were checked. The first is the data source. On 1,440 randomly drawn markets from before

November 2025 (of 1,500 drawn; Appendix D), the price 24 hours before the close rebuilt from Kalshi's hourly histories is within half a cent of the trade-based price for 81% of markets (average difference 0.6 cents), and H1 computed on the same markets is +0.56% with trade prices and +0.78% with hourly prices: the hourly histories do not produce lower returns. The second is the market mix: sports rose from 21% of long shots to 53%, and sports long shots, which were profitable to sell before November 2025, were not afterwards. Re-weighting the later period to the earlier mix of categories raises its return from −0.02% to +0.26%. The break therefore reflects a change in what Kalshi lists, and in how sports long shots are priced, more than a change in the data.

On Polymarket, the strategy loses: −1.56% per dollar, with an interval of [−2.01%, −1.13%]. Long shots priced at 4.0 cents on average won 5.5% of the time. H1 is *Contradicted* on Polymarket, and so is H5. The loss is smaller in plain Yes/No questions and larger in sports and crypto markets named after teams or price moves.

6.3 What a taker would actually pay

Table 7. Selling long shots as a taker at executable prices (Kalshi)

Price paid for "No"	Sample	Mean return	95% interval	Label
100 minus last Yes trade (H1)	All	+0.06%	[−0.05%, 0.17%]	Inconclusive
Half a cent more	All	−0.46%	[−0.57%, −0.35%]	Exploratory
One cent more	All	−0.97%	[−1.08%, −0.86%]	Exploratory
One cent more	Liquid and fresh	−0.11%	[−0.27%, 0.04%]	Exploratory
100 minus last Yes trade	Check sample with a bid, to Nov 2025 (565)	+0.57%	[−1.37%, 2.28%]	Exploratory
The ask (100 minus the Yes bid)	Check sample with a bid, to Nov 2025 (565)	−0.98%	[−2.91%, 0.71%]	Exploratory
100 minus last Yes trade	Check sample with a bid, from Nov 2025 (858)	−0.96%	[−2.78%, 0.62%]	Exploratory
The ask (100 minus the Yes bid)	Check sample with a bid, from Nov 2025 (858)	−1.92%	[−3.73%, −0.33%]	Exploratory

Return per dollar after fees, 24 hours before the close. Liquid and fresh: at least 1,000 contracts traded in the previous 24 hours and a trade in the last 6 hours. Check samples: randomly drawn long-shot markets with a Yes bid at entry, bid and ask recorded at entry (number of markets in brackets); long shots without a bid could not be bought and are left out. Added after the main results were seen. Source: Author's own calculations.

The primary test buys "No" at 100 minus the last traded Yes price. Most trading in long shots is takers buying Yes, so that last trade is often at the Yes ask, and 100 minus it is the price at which a maker would sell "No", not the price a taker would pay. The check samples record the bid and the ask 24 hours before the close. On 60% of long shots before November 2025 and 43% after, nobody was bidding for Yes at all, which means nobody was

offering “No” below $1: the fund could not have bought those contracts. Where there was a bid, the last trade was typically 1 cent above it, and the median spread was 3 and 1 cents in the two periods. Table 7 shows what this costs. Paying half a cent more than the last trade turns H1 into −0.46%, and one cent more into −0.97%; buying at the recorded ask, on the markets where one existed, gives −0.98% and −1.92% in the two check samples, against +0.57% and −0.96% at the last price on the same markets. The +0.06% of H1 is therefore an upper bound: a taker who actually crosses the spread loses money selling long shots on Kalshi, and on many long shots there is nothing to buy.

6.4 Selling long shots as a maker (H2)

Table 8. Selling 1–10 cent long shots as a maker and as a taker, from recorded trades

Sample	Maker	95% interval	Taker	Maker − taker	Label (maker / difference)
Kalshi, all trades to Nov 2025	**+1.89%**	**[1.23%, 2.46%]**	**+0.32%**	**1.57 pp**	**Supported / Supported**
to April 2025	+1.90%	[0.98%, 2.69%]	+0.45%	1.45 pp	Supported / Supported
May – Nov 2025	+1.89%	[1.05%, 2.61%]	+0.22%	1.67 pp	Supported / Supported
trades at least 1 day before close	+2.61%	[1.81%, 3.32%]	+1.45%	1.16 pp	Supported / Supported
Polymarket, all trades	**+0.85%**	**[−0.58%, 1.78%]**	**+0.96%**	**−0.11 pp**	**Inconclusive / Inconclusive**
to April 2025	+0.21%	[−2.37%, 1.79%]	+0.86%	−0.65 pp	Inconclusive / Inconclusive
May 2025 – Jan 2026	+1.72%	[1.17%, 2.17%]	+1.05%	0.67 pp	Supported / Supported
at least 1 day before end date	+0.72%	[−1.06%, 1.88%]	+1.16%	−0.44 pp	Inconclusive / Inconclusive

Return per dollar after fees, weighted by contracts. Maker positions: trades in which a taker bought the long shot. Taker positions: trades at the same prices in which a taker bought the favourite. Both are different trades in the same price band, not the same contracts. Kalshi trade data end on 23 November 2025. Source: Author's own calculations.

Makers did what takers could not. On Kalshi, makers who sold long shots earned +1.89% per dollar (interval [1.23%, 2.46%]), 1.57 percentage points more than takers who bought favourites in the same price band. Both parts of H2 are *Supported*, before and after April 2025 and for trades made at least a day before the close. The maker earns the spread: by posting the price instead of accepting it, the maker sells the long shot at a better price. On Polymarket, the maker return of +0.85% is not significant over the whole period and makers did not beat takers; from May 2025, both are positive and significant.

Table 8 weights trades by contracts, while H1 weights every market equally, and the weighting matters. Weighted by contracts, takers who bought favourites at least a day before the close earned +1.45% on Kalshi; and H1 itself, weighted by the contracts traded in the day before entry, is +1.22% (interval [0.69%, 1.72%], exploratory). Long shots in busy markets were more overpriced than those in quiet ones, so a taker who trades where the volume is earns more than the equal-weighted average suggests, before paying the spread. Figure 2 summarises who collected the house edge.

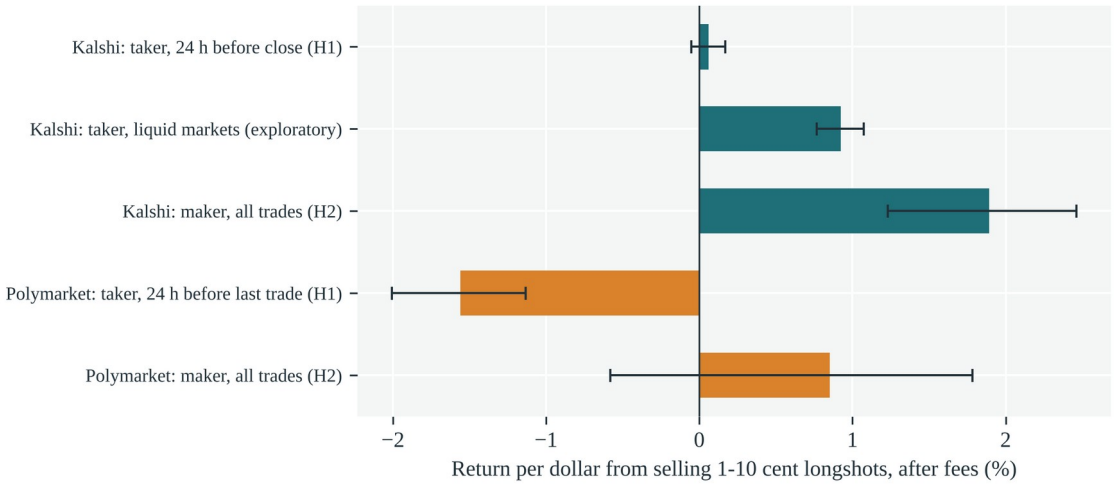


Figure 2. Return per dollar from selling 1–10 cent long shots, after fees, with 95% intervals. Liquid markets: price from a trade in the 6 hours before entry and at least 1,000 contracts traded in the previous 24 hours (exploratory). Taker returns are at the last traded price, an upper bound (Section 6.3). Figure produced using Matplotlib (Hunter, 2007).

6.5 The fund (H3, H4)

Table 9. A $1 million fund selling long shots as a taker

	Kalshi	Polymarket
Period	2021-07 to 2026-10	2023-03 to 2026-01
Final value	$1.456m	$1.181m
Annual return (including interest on cash)	7.39%	5.95%
Annual excess return	3.62%	0.74%
Annualised Sharpe ratio	1.48	0.53
Average share of capital invested	1.1%	0.5%
Positions / turnover	115,534 / $27.1m	21,278 / $5.8m
Profit per dollar traded / per position	0.91% / 0.13%	0.42% / −1.22%
Skewness of monthly returns [95% interval]	2.56 [1.37, 4.72]	−0.02 [−0.50, 0.69]
Worst month / median month	−0.44% / 0.017%	−0.82% / 0.038%
Maximum drawdown	−0.84%	−1.33%
H3 (volatility-seller profile)	Contradicted	Inconclusive
Correlation with the stock market [95% interval]	−0.09 [−0.27, 0.11]	0.26 [0.07, 0.50]
Return in the worst 10% of market months [95% interval]	0.08% [−0.01%, 1.29%]	−0.17% [−0.37%, 0.26%]
H4 (diversifies stocks)	Supported	Inconclusive

Monthly returns in excess of the risk-free rate, to September 2026, the last month of the risk-free data; the fund's cash earns no interest in October 2026. Intervals from a stationary block bootstrap of months (mean block of three months, 10,000 draws). The fund trades at the last traded price, so its returns are upper bounds (Section 6.3). Source: Author's own calculations.

Table 9 and Figure 3 show the fund. On Kalshi, $1 million grew to $1.46 million by October 2026, 7.4% a year including interest, or 3.6% a year above the risk-free rate. The fund earned 0.91% on each dollar it traded, far more than the +0.06% of the average contract, because the volume limit gives it its largest positions in the busiest markets, where long shots were most overpriced (Section 6.4). It trades at the last price, however, and paying the spread would remove most of that margin. It also buys contracts that nobody was offering: in the check samples, 43% to 60% of long shots had no Yes bid 24 hours before the close (Section 6.3), so there was no seller of “No” below $1, and the simulation fills those orders anyway. Its returns and its capacity are therefore both overstated. Even so, it invested only about 1.1% of its capital on an average day, because the markets it trades are small.

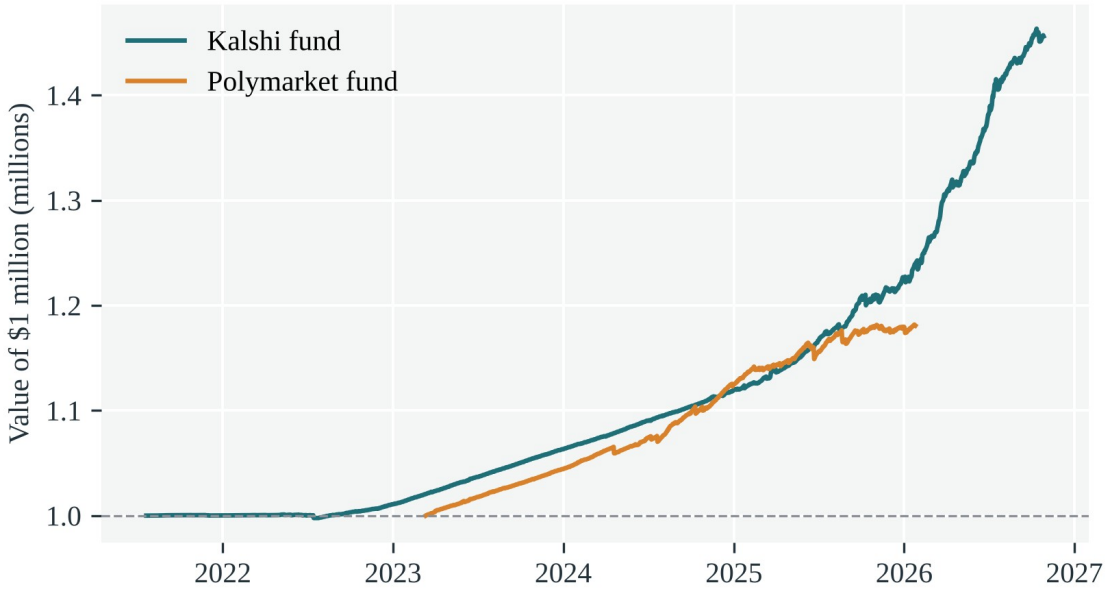


Figure 3. Value of $1 million invested in each fund, with idle cash earning the risk-free rate (to September 2026). Figure produced using Matplotlib (Hunter, 2007).

The Polymarket fund earned 0.42% per dollar traded although every per-contract estimate is negative. The difference is weighting. Counted per position, its trades lost 1.22%; per dollar, they gained, because 98% of its money went to markets with more than $1,000 traded the day before, and within those markets the largest positions did best (+0.44% per dollar against −0.68% per position). The gains also come from 2023 and 2024 (+1.7% and +1.7% per dollar); in 2025 and 2026 the fund lost money on its trades.

The plan's H3 condition is met by Polymarket and not by Kalshi, but neither result says much. Because both funds are almost entirely in cash, their median month is close to zero, so a worst month “six times the median” is almost automatic: the ratios are 26 and 21. The informative statistic is the skewness. Kalshi's is significantly positive (2.56), so H3 is *Contradicted*: with at most 1% of capital in one event and thousands of unrelated positions, no single long shot can hurt the fund, whose worst month lost 0.44%. Polymarket's skewness (−0.02, interval −0.50 to 0.69) is not significantly different from zero. The plan's pass condition is met, but only because the median month is near zero, so H3 is labelled *Inconclusive* (Appendix C): the fund does not have a volatility seller's crash-prone profile either.

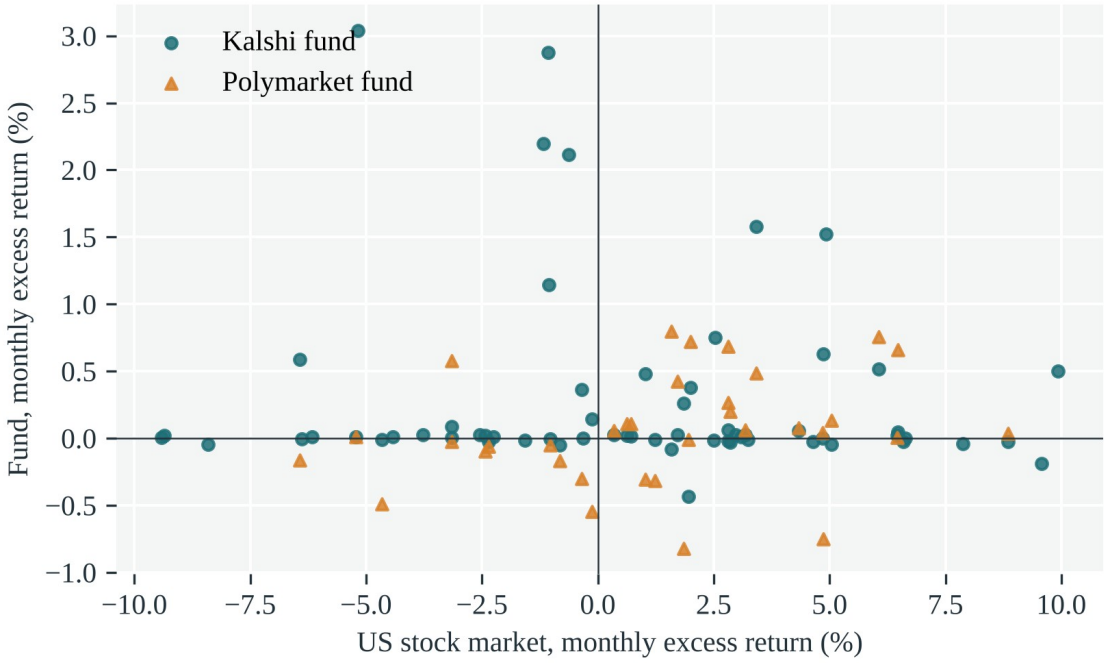


Figure 4. Monthly excess returns of each fund against the US stock market's excess return. Figure produced using Matplotlib (Hunter, 2007).

H4 is *Supported* on Kalshi: the fund's correlation with the stock market is −0.09 (interval −0.27 to 0.11, inside ±0.3) and it gained 0.08% on average in the 7 worst stock-market months, when the market lost 7.3% (Figure 4). That second condition passes on the point estimate only: its interval, −0.01% to 1.29%, just admits a small loss. The stricter rule adopted for H3 is not applied to H4, because the reason for it does not carry over: H3's ratio condition is met almost automatically by a fund that is mostly in cash, whereas H4's conditions are not. Judged by the H3 standard, Kalshi's H4 would be borderline *Inconclusive*; the evidence is that the fund did not lose in bad months, not that it reliably gained. On Polymarket, the correlation of {num(FP['corr'])} meets the condition of being below 0.3, but the fund lost 0.17% in the 4 worst market months, which fails the second condition; with an interval that includes zero, H4 is *Inconclusive* there.

6.6 Robustness and timing

Table 10. Robustness grid, full period: number of specifications by label

Platform and side	Fees	Supported	Inconclusive	Contradicted
Kalshi, taker	in force	7	4	1
Kalshi, taker	none	12	0	0
Kalshi, maker	in force	12	0	0
Kalshi, maker	none	12	0	0
Polymarket, taker	in force	1	2	9
Polymarket, taker	none	1	2	9
Polymarket, maker	in force	0	12	0
Polymarket, maker	none	0	12	0

Each row covers 12 specifications: long-shot thresholds of 1–5, 1–10, 1–15 and 1–20 cents and entry 1 hour, 24 hours or 7 days before the close (for makers, trades made at least 0 hours, 1 day or 7 days before). Taker returns are at the last traded price. Every specification and period is listed in Appendix A. Source: Author's own calculations.

Table 10 summarises the 48 specifications per platform fixed in the plan. The Kalshi maker result holds in 24 of 24. Kalshi takers earn a significant return in 7 of 12 specifications after fees, but these are mostly the entries one hour before the close (+0.81% to +0.90%) and the narrowest threshold; the primary 24-hour entry is the weakest. Polymarket takers lose in most specifications with or without fees, and Polymarket makers are *Inconclusive* in all 24 of their 24 specifications, with and without fees.

Entry times are measured from each market's recorded closing time. A market that closes early because its outcome became known is measured from that early close, which a fund could not have known in advance, and the one-hour entries are the most exposed. Kalshi records an expected expiration for each market; in the check samples, the median market stopped trading 4 hours before it, and 22% stopped more than a day before, which marks an early close. Leaving those out lowers the 24-hour return in the check samples from +0.42% to +0.26%, so early closes flatter H1 somewhat. The one-hour return is much the same without them (+1.30% against +1.26%), so the one-hour results are not driven by early closes; they are, however, measured at the last traded price, and the spread argument of Section 6.3 applies to them in full.

6.7 Liquidity and stale prices (exploratory)

Table 11. Selling long shots as a taker, by liquidity (exploratory)

Sample	Contracts	Mean return	95% interval
All long shots (H1)	148,939	+0.06%	[−0.05%, 0.17%]
Price from a trade in the 6 hours before entry	122,666	−0.05%	[−0.18%, 0.08%]
Price older than 6 hours	26,273	+0.56%	[0.39%, 0.72%]
At least 1,000 contracts traded in the previous 24 hours	49,459	+0.89%	[0.75%, 1.04%]
and price from a trade in the last 6 hours	45,730	+0.92%	[0.77%, 1.07%]
Fewer than 100 contracts traded in the previous 24 hours	55,894	−0.90%	[−1.10%, −0.70%]
Polymarket: at least $1,000 traded in the previous 24 hours	6,922	−0.53%	[−1.25%, 0.13%]
Polymarket: less than $10 traded	7,378	−3.79%	[−4.75%, −2.96%]

Not part of the plan: added after the main results were seen. Kalshi rows: July 2021 to October 2026, 24 hours before the close, at the last traded price. Source: Author's own calculations.

These splits were not planned. In markets where at least 1,000 contracts traded the day before and the price came from a recent trade, selling long shots at the last price earned +0.92% per dollar; paying one cent more turns this into −0.11%. In thin markets, with fewer than 100 contracts traded, it lost 0.90%. Prices older than six hours look profitable (+0.56%), but that is very likely a mirage: they belong to markets that had stopped trading, often because the outcome was nearly settled. In the check sample before November 2025, only 18% of long shots with a price older than six hours had a Yes bid 24 hours before the close (780 markets), against 64% of those with a recent price; for most stale-price long shots there was no seller of "No" below $1, and no taker could have bought at the last price. On Polymarket, long shots in markets with almost no trading won far more often than their price said. The edge, where it exists, is small, lives in busy markets, and is mostly the spread.

7. Results: Prediction Markets as Crash Insurance

7.1 The cost of insurance (HB1, HB5)

Table 12. Crash insurance from S&P 500 and Nasdaq-100 range contracts

Prices	Test	Estimate	95% interval	p (Holm)	Label
Pre-registered prices	HB1	−69.9%	[−82.8%, −54.0%]	<0.001	Supported
	HB2	−0.34	[−0.49, −0.18]	<0.001	Supported
	HB3	−0.03%	[−0.23%, 0.24%]	1.000	Inconclusive
	HB4	−0.213	[−0.309, −0.137]	1.000	Contradicted
	HB5	−78.3%	[−90.8%, −63.4%]	<0.001	Supported
Prices from trades in the 24 h before entry	HB1	−76.2%	[−89.5%, −60.8%]	<0.001	Exploratory
	HB2	−0.21	[−0.38, −0.03]	0.035	Exploratory
	HB3	−0.06%	[−0.23%, 0.21%]	1.000	Exploratory
	HB4	−0.174	[−0.258, −0.113]	1.000	Exploratory
	HB5	−75.4%	[−91.0%, −57.0%]	<0.001	Exploratory
Traded prices adding up to 0.8–1.2	HB1	−62.0%	[−91.4%, −31.8%]	<0.001	Exploratory
	HB2	−0.29	[−0.53, −0.03]	0.044	Exploratory
	HB3	−0.02%	[−0.19%, 0.15%]	1.000	Exploratory
	HB4	−0.113	[−0.178, −0.057]	1.000	Exploratory
	HB5	−59.3%	[−100.0%, −21.0%]	0.008	Exploratory

Entry 24 hours before the close, insurance from 1% below the market's median forecast, 0.01 bundles per dollar of stocks. HB1 and HB5: return per dollar of insurance; HB2: correlation of the insurance's daily profit with the market; HB3: change in the 5% expected shortfall of daily returns (positive = lower tail risk); HB4: change in the daily Sharpe ratio. Stationary block bootstrap over dates (mean block of 5 days, 10,000 draws); an S&P 500 and a Nasdaq-100 bundle on the same date are always resampled together. The pre-registered version is the primary result; the other two were added after it was seen. Source: Author's own calculations.

Table 12 reports Part B. With prices as specified in the plan, the fund could insure 291 index-days (165 for the S&P 500 and 126 for the Nasdaq-100) on 182 dates between 2022-04 and 2026-10. A bundle paying $1 cost 28.5 cents on average and paid out on 8.6% of index-days. Its buyers lost 69.9% per dollar (interval −82.8% to −54.0%): HB1 is *Supported*, and so is HB5 after April 2025 (−78.3%). At 0.01 bundles per dollar of stocks, the insurance cost 0.285% of the portfolio per insured index-day (twice that on a date when both indexes are insured); bought on one index every trading day, that would be about 72% of the portfolio a year, or a net loss of about 50% a year after payouts.

A loss this large invites suspicion, and many range prices 24 hours before the close are indeed stale: on some days most ranges show the same price, left over from the contract's first

trades, and the prices of a day's ranges add up to as much as $6.92 instead of about $1. Two stricter versions use a range's price only if it traded in the 24 hours before entry, and in addition only on days when those prices add up to between 0.8 and 1.2. Buyers still lose 76% and 62% per dollar. Figure 5 shows the same picture at every entry time from November 2025: the 24-hour bar uses only the 39% of days on which some ranges were listed early enough. The cause is not Kalshi's minimum price of one cent: ranges far below the forecast traded at 3 to 17 cents but won only 1% to 3% of the time.

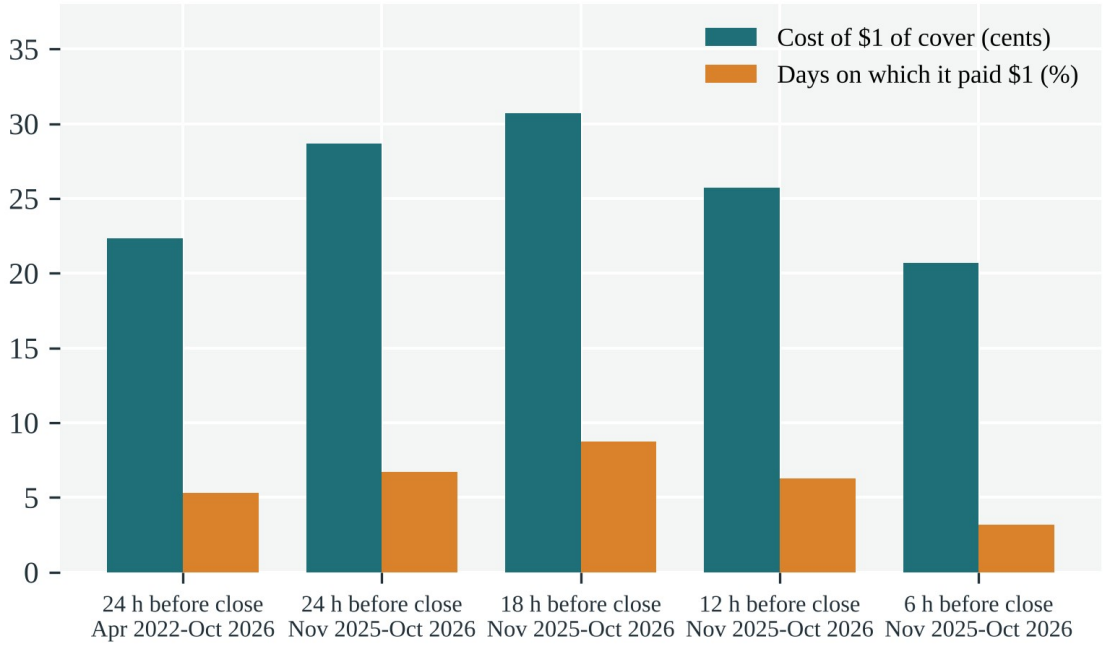


Figure 5. Average cost of a bundle paying $1 if the index closes more than 1% below the market's median forecast, and the share of days on which it paid, by entry time. Prices from trades in the 24 hours before entry. Entries other than 24 hours exist only from November 2025, and 24-hour entries in that period only on days listed early. Figure produced using Matplotlib (Hunter, 2007).

7.2 Does it protect the portfolio? (HB2–HB4)

Table 13. The insurance on the 15 worst market days among insured dates

Date	US market	Bundles bought	Bundles that paid	Cost of bundles (c)
2025-04-04	−5.90%	1	1	5.3
2022-04-29	−3.54%	1	0	19.2
2022-05-09	−3.52%	1	0	12.7
2022-04-26	−2.99%	1	0	2.1
2026-06-05	−2.74%	2	0	66.0
2024-08-02	−2.09%	1	0	11.7
2025-03-28	−2.05%	1	1	37.0
2025-02-21	−1.88%	2	0	14.9
2024-09-06	−1.80%	2	2	34.2
2026-03-27	−1.73%	2	2	55.5
2025-08-01	−1.72%	1	0	31.5
2026-03-20	−1.56%	2	2	79.0
2024-04-12	−1.51%	2	1	56.7
2024-11-15	−1.36%	2	2	31.0
2026-03-06	−1.35%	2	1	85.3

One bundle per index (S&P 500, Nasdaq-100) with prices 24 hours before the close; pre-registered prices. US market: Kenneth French's market return that day. Source: Author's own calculations.

The insurance does pay when stocks fall: its daily profit has a correlation of −0.34 with the market's return, and HB2 is *Supported* with the pre-registered prices (with traded prices only, the correlation is −0.21). But it often missed the days that mattered. Table 13 lists the 15 worst market days on which the fund was insured: it paid on 8 of them, and on only 39% of all insured days when the market fell more than 1%. The bundle pays when the index closes 1% below the market's own forecast, which was often set from stale prices and need not match the previous close; and the Nasdaq-100 and S&P 500 do not always fall with the broad market. As a result, the 5% expected shortfall of the portfolio's daily returns is essentially unchanged (a difference of −0.03 points, HB3 *Inconclusive*), and the cost lowers the daily Sharpe ratio by 0.21 (HB4 *Contradicted*).

7.3 Robustness

Appendix B reports 216 combinations of entry time, insurance level *d*, insurance amount *h*, index and fees. In all of the 72 combinations of entry, level, index and fees, the buyers' loss is significant, and the return per dollar ranges from −100% to −50%. The insurance's profit is negatively correlated with the market, with an interval below zero, in 58

of those 72. Across all 216 combinations, the reduction in tail risk is significant in none, and the Sharpe ratio falls significantly in all.

7.4 Fed decisions and inflation (exploratory)

Table 14. Fed decisions and inflation releases on Kalshi (descriptive)

	Fed decisions	CPI releases
Events	22	63
Period	2023-05 to 2026-09	2021-07 to 2026-10
Probability given to the actual outcome a day before	92%	–
Average surprise	1.5 bp	0.80 thresholds
Average absolute market move that day	0.74%	0.95%
Correlation of surprise with the market that day	0.11	−0.40

Surprise: the outcome minus the contracts' expectation 24 hours before (Fed: change in basis points; CPI: number of 0.1-point thresholds exceeded). Average absolute market move on all days: 0.79%. Not tested; too few events. Source: Author's own calculations.

Kalshi's Fed-decision markets gave the actual outcome an average probability of 92% a day ahead, leaving little surprise and no visible link to the stock market on decision days. Inflation releases are different: the market moved 0.95% on CPI days against 0.79% on an average day, and an inflation figure hotter than the contracts expected went with lower stock prices (correlation −0.40). A contract paying on high inflation therefore behaves like a hedge for CPI days. With 63 releases, this is a description, not a test.

8. DISCUSSION

8.1 Where the house edge goes

The longshot bias on prediction markets is real, but it is not a free lunch for whoever takes the other side. On Kalshi, it is concentrated in the final hours, when many outcomes are known, and in the spread between the prices makers post and takers accept. A day before the close, long shots are only mildly overpriced, and the fee and the spread absorb what is left. The “house” on a prediction market is therefore not a fund that sells every long shot, but a market maker who posts prices and lets impatient buyers come to it. That fits the account of Bürgi et al. (2026), in which the bias comes from takers, and it is consistent with Moskowitz (2021): in betting markets, anomalies tend to be smaller than the cost of trading on them.

Polymarket shows that the bias is not universal. Its long shots are, if anything, underpriced a day before the end, especially in fast sports and crypto markets where prices go stale between trades. The two platforms attract different traders, charge different fees and list different questions; a pattern that holds on one cannot be assumed on the other.

8.2 Capacity

The fund's most striking statistic is not its return but its use of capital: about 1% on an average day. Whatever edge exists lives in busy markets, but even those trade only thousands of contracts a day, and a fund cannot buy more than a fraction of them without moving the price. A $1 million fund would sit mostly in cash; a $100 million fund would find nothing to do. Selling long shots on prediction markets may suit a market maker with a small book, but not a fund that has to deploy investors' money.

8.3 Insurance for gamblers, and for funds

Part B turns the question around. A stock fund that wants protection against a bad day is in the position of the longshot buyer, and it pays accordingly. Kalshi's index ranges price a fall of more than 1% several times above how often it happened, much as S&P 500 puts are priced above their average payout (Coval & Shumway, 2001; Bondarenko, 2014). The insurance pays when stocks fall, but too rarely and too expensively to improve the portfolio, a result that echoes Israelov's (2019) finding for protective puts. The markets are young, the index ranges thin, and their prices a day ahead often stale; for now, a fund is better off on the other side of the trade.

Taken together, the two parts tell one story. Whoever buys the unlikely outcome, a bettor on an outsider or a fund insuring against a crash, overpays; whoever sells it is paid. The house edge exists without a house. But collecting it requires being the patient side of the trade, in markets with real trading, and in amounts small enough not to move the price.

9. Limitations and Conclusion

9.1 Limitations

- **Prices and fills.** The taker tests assume the fund could buy at the last traded price, which is often the price a maker would get; Section 6.3 shows that paying the spread turns the taker return negative, so H1 and the fund's returns are upper bounds. The maker tests use trades that happened but cannot say whether the fund's own orders would have been filled ahead of others'.
- **Data after November 2025.** Kalshi prices after 23 November 2025 come from hourly price histories rather than individual trades, and the maker test on Kalshi stops on that date. Polymarket's data end on 25 January 2026.
- **Fees.** Kalshi's current series multipliers are applied to past trades. Fees are computed on 100-contract orders; smaller orders pay more per contract, and the validation suggests the published benchmark used the costlier rule.
- **Clustering on Polymarket.** Without event identifiers, Polymarket contracts are grouped by day, which is coarser than events and makes the intervals wider.
- **Part B.** Only days with prices 24 hours before the close can be insured, and many of those prices are stale. There are no S&P 500 option prices in the data, so the insurance cannot be compared directly with puts.
- **Timing.** “Twenty-four hours before the close” uses each market's recorded closing time; markets that closed early when their outcome became known are measured from that time, which a fund could not have known in advance. Section 6.6 measures this on a sample; the full sample could not be re-run on scheduled times.
- **Regulation.** The study measures returns, not whether a fund could legally trade these contracts from a given country.
- **The plan's timeline.** The plan is a dated document with a version history, but it is not lodged in a public registry; the plan (27 October), Part B (30 October) and this paper (first draft 30 October, revised 31 October) are close together, and readers must take the order of events in Appendix C on trust.

9.2 Conclusion

This paper asked whether a fund could act as the house on prediction markets by selling the long shots that bettors overpay for, and whether a stock fund could buy crash

insurance from the same markets. On Kalshi, selling long shots as a taker a day before the close earns at most +0.06% per dollar after fees, and loses money once the spread is paid; on Polymarket, it loses 1.56%. The premium is real but goes to makers, who earned +1.89% on Kalshi. Busy markets show a taker edge at the last traded price, but it is mostly the spread and too small to employ much capital. A fund built on it would be safe and uncorrelated with stocks, but mostly in cash.

For a stock fund, the same markets sell protection at a steep price: one-day crash insurance on the S&P 500 and Nasdaq-100 lost its buyers 70% per dollar, paid out when stocks fell, and still left the portfolio worse off. Prediction markets reward the patient side of every trade. A fund can be that side, but on today's markets, only at the margin.

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APPENDIX A. ROBUSTNESS GRID, PART A

Every specification of the pre-registered grid, full period. Kalshi trade data for the maker side end on 23 November 2025; Kalshi's 1-hour taker entry uses the trade data only.

Table 15. Robustness grid, Part A, full period

Platform	Side	Threshold	Entry	Fees	Mean	95% interval	p	Label
Kalshi	maker	1-10 c	1 hour	in force	+1.89%	[1.23%, 2.46%]	<0.001	Supported
Kalshi	maker	1-10 c	24 hours	in force	+2.61%	[1.81%, 3.32%]	<0.001	Supported
Kalshi	maker	1-10 c	7 days	in force	+2.87%	[1.79%, 3.82%]	<0.001	Supported
Kalshi	maker	1-15 c	1 hour	in force	+2.08%	[1.21%, 2.86%]	<0.001	Supported
Kalshi	maker	1-15 c	24 hours	in force	+3.23%	[2.23%, 4.13%]	<0.001	Supported
Kalshi	maker	1-15 c	7 days	in force	+3.73%	[2.42%, 4.85%]	<0.001	Supported
Kalshi	maker	1-20 c	1 hour	in force	+2.08%	[0.98%, 3.06%]	<0.001	Supported
Kalshi	maker	1-20 c	24 hours	in force	+3.46%	[2.25%, 4.59%]	<0.001	Supported
Kalshi	maker	1-20 c	7 days	in force	+4.10%	[2.50%, 5.56%]	<0.001	Supported
Kalshi	maker	1-5 c	1 hour	in force	+1.38%	[0.96%, 1.72%]	<0.001	Supported
Kalshi	maker	1-5 c	24 hours	in force	+2.02%	[1.58%, 2.36%]	<0.001	Supported
Kalshi	maker	1-5 c	7 days	in force	+2.23%	[1.64%, 2.64%]	<0.001	Supported
Kalshi	maker	1-10 c	1 hour	none	+1.95%	[1.28%, 2.51%]	<0.001	Supported
Kalshi	maker	1-10 c	24 hours	none	+2.64%	[1.83%, 3.37%]	<0.001	Supported
Kalshi	maker	1-10 c	7 days	none	+2.89%	[1.81%, 3.88%]	<0.001	Supported
Kalshi	maker	1-15 c	1 hour	none	+2.16%	[1.28%, 2.94%]	<0.001	Supported
Kalshi	maker	1-15 c	24 hours	none	+3.27%	[2.26%, 4.17%]	<0.001	Supported
Kalshi	maker	1-15 c	7 days	none	+3.76%	[2.45%, 4.88%]	<0.001	Supported
Kalshi	maker	1-20 c	1 hour	none	+2.19%	[1.08%, 3.17%]	<0.001	Supported
Kalshi	maker	1-20 c	24 hours	none	+3.51%	[2.30%, 4.64%]	<0.001	Supported
Kalshi	maker	1-20 c	7 days	none	+4.13%	[2.53%, 5.61%]	<0.001	Supported
Kalshi	maker	1-5 c	1 hour	none	+1.41%	[0.99%, 1.75%]	<0.001	Supported
Kalshi	maker	1-5 c	24 hours	none	+2.04%	[1.59%, 2.38%]	<0.001	Supported
Kalshi	maker	1-5 c	7 days	none	+2.24%	[1.65%, 2.66%]	<0.001	Supported
Kalshi	taker	1-10 c	1 hour	in force	+0.85%	[0.74%, 0.96%]	<0.001	Supported
Kalshi	taker	1-10 c	24 hours	in force	+0.06%	[−0.05%, 0.17%]	0.143	Inconclusive
Kalshi	taker	1-10 c	7 days	in force	+0.27%	[0.03%, 0.50%]	0.015	Supported
Kalshi	taker	1-15 c	1 hour	in force	+0.87%	[0.73%, 1.00%]	<0.001	Supported
Kalshi	taker	1-15 c	24 hours	in force	−0.11%	[−0.24%, 0.01%]	0.966	Inconclusive
Kalshi	taker	1-15 c	7 days	in force	+0.20%	[−0.07%, 0.48%]	0.072	Inconclusive
Kalshi	taker	1-20 c	1 hour	in force	+0.90%	[0.74%, 1.05%]	<0.001	Supported
Kalshi	taker	1-20 c	24 hours	in force	−0.31%	[−0.46%, −0.18%]	1.000	Contradicted
Kalshi	taker	1-20 c	7 days	in force	+0.10%	[−0.22%, 0.41%]	0.264	Inconclusive
Kalshi	taker	1-5 c	1 hour	in force	+0.81%	[0.72%, 0.89%]	<0.001	Supported
Kalshi	taker	1-5 c	24 hours	in force	+0.20%	[0.11%, 0.29%]	<0.001	Supported
Kalshi	taker	1-5 c	7 days	in force	+0.38%	[0.19%, 0.57%]	<0.001	Supported
Kalshi	taker	1-10 c	1 hour	none	+1.02%	[0.91%, 1.13%]	<0.001	Supported
Kalshi	taker	1-10 c	24 hours	none	+0.33%	[0.22%, 0.44%]	<0.001	Supported
Kalshi	taker	1-10 c	7 days	none	+0.51%	[0.26%, 0.74%]	<0.001	Supported
Kalshi	taker	1-15 c	1 hour	none	+1.08%	[0.95%, 1.22%]	<0.001	Supported
Kalshi	taker	1-15 c	24 hours	none	+0.27%	[0.14%, 0.40%]	<0.001	Supported

Platform	Side	Threshold	Entry	Fees	Mean	95% interval	p	Label
Kalshi	taker	1-15 c	7 days	none	+0.52%	[0.24%, 0.80%]	<0.001	Supported
Kalshi	taker	1-20 c	1 hour	none	+1.17%	[1.01%, 1.32%]	<0.001	Supported
Kalshi	taker	1-20 c	24 hours	none	+0.19%	[0.05%, 0.33%]	0.005	Supported
Kalshi	taker	1-20 c	7 days	none	+0.51%	[0.19%, 0.82%]	<0.001	Supported
Kalshi	taker	1-5 c	1 hour	none	+0.93%	[0.84%, 1.01%]	<0.001	Supported
Kalshi	taker	1-5 c	24 hours	none	+0.36%	[0.27%, 0.44%]	<0.001	Supported
Kalshi	taker	1-5 c	7 days	none	+0.53%	[0.33%, 0.71%]	<0.001	Supported
Polymarket	maker	1-10 c	1 hour	in force	+0.71%	[−0.93%, 1.78%]	0.155	Inconclusive
Polymarket	maker	1-10 c	24 hours	in force	+0.72%	[−1.06%, 1.88%]	0.178	Inconclusive
Polymarket	maker	1-10 c	7 days	in force	+0.53%	[−1.60%, 1.89%]	0.261	Inconclusive
Polymarket	maker	1-15 c	1 hour	in force	+0.93%	[−0.59%, 1.98%]	0.094	Inconclusive
Polymarket	maker	1-15 c	24 hours	in force	+0.98%	[−0.76%, 2.14%]	0.111	Inconclusive
Polymarket	maker	1-15 c	7 days	in force	+0.77%	[−1.30%, 2.14%]	0.186	Inconclusive
Polymarket	maker	1-20 c	1 hour	in force	+1.01%	[−0.46%, 2.07%]	0.076	Inconclusive
Polymarket	maker	1-20 c	24 hours	in force	+1.12%	[−0.60%, 2.34%]	0.084	Inconclusive
Polymarket	maker	1-20 c	7 days	in force	+0.92%	[−1.21%, 2.35%]	0.157	Inconclusive
Polymarket	maker	1-5 c	1 hour	in force	+0.25%	[−1.63%, 1.39%]	0.347	Inconclusive
Polymarket	maker	1-5 c	24 hours	in force	+0.23%	[−1.86%, 1.47%]	0.362	Inconclusive
Polymarket	maker	1-5 c	7 days	in force	+0.07%	[−2.34%, 1.48%]	0.433	Inconclusive
Polymarket	maker	1-10 c	1 hour	none	+0.72%	[−0.93%, 1.78%]	0.154	Inconclusive
Polymarket	maker	1-10 c	24 hours	none	+0.72%	[−1.06%, 1.88%]	0.178	Inconclusive
Polymarket	maker	1-10 c	7 days	none	+0.53%	[−1.60%, 1.89%]	0.261	Inconclusive
Polymarket	maker	1-15 c	1 hour	none	+0.94%	[−0.58%, 1.99%]	0.092	Inconclusive
Polymarket	maker	1-15 c	24 hours	none	+0.98%	[−0.76%, 2.14%]	0.111	Inconclusive
Polymarket	maker	1-15 c	7 days	none	+0.77%	[−1.30%, 2.14%]	0.186	Inconclusive
Polymarket	maker	1-20 c	1 hour	none	+1.02%	[−0.46%, 2.08%]	0.074	Inconclusive
Polymarket	maker	1-20 c	24 hours	none	+1.12%	[−0.60%, 2.34%]	0.084	Inconclusive
Polymarket	maker	1-20 c	7 days	none	+0.92%	[−1.21%, 2.35%]	0.157	Inconclusive
Polymarket	maker	1-5 c	1 hour	none	+0.25%	[−1.63%, 1.39%]	0.347	Inconclusive
Polymarket	maker	1-5 c	24 hours	none	+0.23%	[−1.86%, 1.47%]	0.362	Inconclusive
Polymarket	maker	1-5 c	7 days	none	+0.07%	[−2.34%, 1.48%]	0.433	Inconclusive
Polymarket	taker	1-10 c	1 hour	in force	−0.07%	[−0.36%, 0.22%]	0.681	Inconclusive
Polymarket	taker	1-10 c	24 hours	in force	−1.56%	[−2.01%, −1.13%]	1.000	Contradicted
Polymarket	taker	1-10 c	7 days	in force	−2.02%	[−2.79%, −1.32%]	1.000	Contradicted
Polymarket	taker	1-15 c	1 hour	in force	+0.24%	[−0.07%, 0.56%]	0.067	Inconclusive
Polymarket	taker	1-15 c	24 hours	in force	−1.35%	[−1.81%, −0.91%]	1.000	Contradicted
Polymarket	taker	1-15 c	7 days	in force	−2.04%	[−2.83%, −1.30%]	1.000	Contradicted
Polymarket	taker	1-20 c	1 hour	in force	+0.77%	[0.44%, 1.11%]	<0.001	Supported
Polymarket	taker	1-20 c	24 hours	in force	−1.33%	[−1.78%, −0.87%]	1.000	Contradicted
Polymarket	taker	1-20 c	7 days	in force	−2.15%	[−2.95%, −1.41%]	1.000	Contradicted
Polymarket	taker	1-5 c	1 hour	in force	−0.45%	[−0.72%, −0.19%]	1.000	Contradicted
Polymarket	taker	1-5 c	24 hours	in force	−1.32%	[−1.76%, −0.91%]	1.000	Contradicted
Polymarket	taker	1-5 c	7 days	in force	−1.41%	[−2.14%, −0.78%]	1.000	Contradicted
Polymarket	taker	1-10 c	1 hour	none	−0.07%	[−0.36%, 0.23%]	0.676	Inconclusive
Polymarket	taker	1-10 c	24 hours	none	−1.56%	[−2.01%, −1.13%]	1.000	Contradicted
Polymarket	taker	1-10 c	7 days	none	−2.01%	[−2.79%, −1.31%]	1.000	Contradicted
Polymarket	taker	1-15 c	1 hour	none	+0.25%	[−0.07%, 0.56%]	0.065	Inconclusive
Polymarket	taker	1-15 c	24 hours	none	−1.35%	[−1.80%, −0.91%]	1.000	Contradicted
Polymarket	taker	1-15 c	7 days	none	−2.03%	[−2.83%, −1.30%]	1.000	Contradicted
Polymarket	taker	1-20 c	1 hour	none	+0.77%	[0.44%, 1.11%]	<0.001	Supported

Platform	Side	Threshold	Entry	Fees	Mean	95% interval	p	Label
Polymarket	taker	1-20 c	24 hours	none	−1.32%	[−1.78%, −0.87%]	1.000	Contradicted
Polymarket	taker	1-20 c	7 days	none	−2.15%	[−2.95%, −1.41%]	1.000	Contradicted
Polymarket	taker	1-5 c	1 hour	none	−0.45%	[−0.72%, −0.19%]	1.000	Contradicted
Polymarket	taker	1-5 c	24 hours	none	−1.32%	[−1.76%, −0.91%]	1.000	Contradicted
Polymarket	taker	1-5 c	7 days	none	−1.41%	[−2.14%, −0.78%]	1.000	Contradicted

Return per dollar. Taker: price snapshots; maker: recorded trades made at least 0 hours, 1 day or 7 days before the close. Source: Author's own calculations.

Appendix B. Robustness Grid, Part B

Every combination of entry time, insurance level d, index and fees, with $h = 0.01$. Entries other than 24 hours exist only from November 2025.

Table 16. Robustness grid, Part B

Entry	d	Index	Fees	Index-days	Cost (c)	Paid	HB1	HB2	HB4
24 h	0.5%	Nasdaq-100	in force	141	35.9	17.0%	−53%	−0.33	−0.209
24 h	0.5%	Nasdaq-100	none	141	33.8	17.0%	−50%	−0.34	−0.186
24 h	0.5%	S&P 500	in force	191	35.9	11.0%	−69%	−0.42	−0.259
24 h	0.5%	S&P 500	none	191	33.8	11.0%	−67%	−0.43	−0.237
24 h	0.5%	pooled	in force	332	35.9	13.6%	−62%	−0.37	−0.238
24 h	0.5%	pooled	none	332	33.8	13.6%	−60%	−0.38	−0.215
24 h	1.0%	Nasdaq-100	in force	126	29.8	11.1%	−63%	−0.25	−0.210
24 h	1.0%	Nasdaq-100	none	126	28.1	11.1%	−60%	−0.25	−0.192
24 h	1.0%	S&P 500	in force	165	27.6	6.7%	−76%	−0.43	−0.214
24 h	1.0%	S&P 500	none	165	25.9	6.7%	−74%	−0.44	−0.196
24 h	1.0%	pooled	in force	291	28.5	8.6%	−70%	−0.34	−0.213
24 h	1.0%	pooled	none	291	26.8	8.6%	−68%	−0.35	−0.195
24 h	2.0%	Nasdaq-100	in force	103	21.6	4.9%	−77%	−0.24	−0.197
24 h	2.0%	Nasdaq-100	none	103	20.3	4.9%	−76%	−0.24	−0.183
24 h	2.0%	S&P 500	in force	125	17.7	3.2%	−82%	−0.44	−0.145
24 h	2.0%	S&P 500	none	125	16.6	3.2%	−81%	−0.45	−0.134
24 h	2.0%	pooled	in force	228	19.4	3.9%	−80%	−0.34	−0.168
24 h	2.0%	pooled	none	228	18.3	3.9%	−78%	−0.35	−0.156
18 h	0.5%	Nasdaq-100	in force	69	54.8	15.9%	−71%	−0.22	−0.474
18 h	0.5%	Nasdaq-100	none	69	52.1	15.9%	−69%	−0.23	−0.445
18 h	0.5%	S&P 500	in force	81	55.5	14.8%	−73%	−0.48	−0.581
18 h	0.5%	S&P 500	none	81	52.4	14.8%	−72%	−0.48	−0.538
18 h	0.5%	pooled	in force	150	55.2	15.3%	−72%	−0.34	−0.528
18 h	0.5%	pooled	none	150	52.3	15.3%	−71%	−0.35	−0.493
18 h	1.0%	Nasdaq-100	in force	68	43.3	11.8%	−73%	−0.18	−0.390
18 h	1.0%	Nasdaq-100	none	68	41.1	11.8%	−71%	−0.19	−0.367
18 h	1.0%	S&P 500	in force	80	37.9	6.2%	−84%	−0.47	−0.434
18 h	1.0%	S&P 500	none	80	35.7	6.2%	−82%	−0.48	−0.403
18 h	1.0%	pooled	in force	148	40.4	8.8%	−78%	−0.30	−0.415
18 h	1.0%	pooled	none	148	38.2	8.8%	−77%	−0.31	−0.388
18 h	2.0%	Nasdaq-100	in force	61	29.5	3.3%	−89%	−0.24	−0.335
18 h	2.0%	Nasdaq-100	none	61	28.0	3.3%	−88%	−0.25	−0.316
18 h	2.0%	S&P 500	in force	72	22.0	0.0%	−100%	−0.21	−0.269
18 h	2.0%	S&P 500	none	72	20.8	0.0%	−100%	−0.21	−0.253
18 h	2.0%	pooled	in force	133	25.5	1.5%	−94%	−0.21	−0.299
18 h	2.0%	pooled	none	133	24.1	1.5%	−94%	−0.22	−0.282
12 h	0.5%	Nasdaq-100	in force	100	50.3	18.0%	−64%	−0.22	−0.363
12 h	0.5%	Nasdaq-100	none	100	47.9	18.0%	−62%	−0.23	−0.340
12 h	0.5%	S&P 500	in force	150	40.3	11.3%	−72%	−0.23	−0.333
12 h	0.5%	S&P 500	none	150	38.1	11.3%	−70%	−0.23	−0.308

Entry	d	Index	Fees	Index-days	Cost (c)	Paid	HB1	HB2	HB4
12 h	0.5%	pooled	in force	250	44.3	14.0%	−68%	−0.22	−0.346
12 h	0.5%	pooled	none	250	42.0	14.0%	−67%	−0.22	−0.322
12 h	1.0%	Nasdaq-100	in force	92	39.3	14.1%	−64%	−0.22	−0.293
12 h	1.0%	Nasdaq-100	none	92	37.3	14.1%	−62%	−0.23	−0.273
12 h	1.0%	S&P 500	in force	132	29.9	4.5%	−85%	−0.23	−0.293
12 h	1.0%	S&P 500	none	132	28.1	4.5%	−84%	−0.23	−0.273
12 h	1.0%	pooled	in force	224	33.7	8.5%	−75%	−0.21	−0.294
12 h	1.0%	pooled	none	224	31.9	8.5%	−73%	−0.22	−0.275
12 h	2.0%	Nasdaq-100	in force	79	29.3	3.8%	−87%	−0.10	−0.310
12 h	2.0%	Nasdaq-100	none	79	27.8	3.8%	−86%	−0.11	−0.294
12 h	2.0%	S&P 500	in force	115	19.2	0.9%	−95%	−0.20	−0.210
12 h	2.0%	S&P 500	none	115	18.1	0.9%	−95%	−0.20	−0.197
12 h	2.0%	pooled	in force	194	23.3	2.1%	−91%	−0.14	−0.251
12 h	2.0%	pooled	none	194	22.1	2.1%	−91%	−0.14	−0.238
6 h	0.5%	Nasdaq-100	in force	177	36.2	14.1%	−61%	−0.29	−0.268
6 h	0.5%	Nasdaq-100	none	177	34.1	14.1%	−59%	−0.29	−0.244
6 h	0.5%	S&P 500	in force	204	32.4	9.3%	−71%	−0.25	−0.285
6 h	0.5%	S&P 500	none	204	30.5	9.3%	−69%	−0.26	−0.263
6 h	0.5%	pooled	in force	381	34.1	11.5%	−66%	−0.27	−0.277
6 h	0.5%	pooled	none	381	32.2	11.5%	−64%	−0.27	−0.254
6 h	1.0%	Nasdaq-100	in force	156	27.2	5.8%	−79%	−0.14	−0.249
6 h	1.0%	Nasdaq-100	none	156	25.7	5.8%	−78%	−0.15	−0.232
6 h	1.0%	S&P 500	in force	169	25.3	1.8%	−93%	−0.20	−0.277
6 h	1.0%	S&P 500	none	169	23.8	1.8%	−93%	−0.21	−0.260
6 h	1.0%	pooled	in force	325	26.2	3.7%	−86%	−0.17	−0.263
6 h	1.0%	pooled	none	325	24.7	3.7%	−85%	−0.17	−0.247
6 h	2.0%	Nasdaq-100	in force	118	19.5	3.4%	−83%	−0.12	−0.190
6 h	2.0%	Nasdaq-100	none	118	18.4	3.4%	−82%	−0.12	−0.178
6 h	2.0%	S&P 500	in force	128	19.0	0.0%	−100%	−0.07	−0.205
6 h	2.0%	S&P 500	none	128	18.0	0.0%	−100%	−0.07	−0.194
6 h	2.0%	pooled	in force	246	19.3	1.6%	−92%	−0.10	−0.198
6 h	2.0%	pooled	none	246	18.2	1.6%	−91%	−0.10	−0.187

HB1: return per dollar of insurance; HB2: correlation with the market; HB4: change in the daily Sharpe ratio. Source: Author's own calculations.

APPENDIX C. CHANGES TO THE PLAN

The plan was fixed on 27 October 2026 and Part B was added on 30 October 2026. The changes below were recorded as they were made, with whether any result had been seen.

1. **Sample end (28 and 30 October 2026).** The plan expected Kalshi data to the latest settled market; a first attempt ended the sample on 23 November 2025, and the extension to 27 October 2026 was completed before any result was computed.
2. **Volume threshold (28 October).** "Traded at least $1,000" is measured as at least 1,000 contracts, which matches the published sample most closely. Seen before the choice: the validation (V) under both definitions; no other result.
3. **Prices after November 2025 (30 October).** Hourly price histories replace individual trades, because downloading every trade would have taken about a week; the Kalshi maker test therefore ends on 23 November 2025. No result had been seen.
4. **Combination contracts (30 October).** Multi-leg contracts are excluded after November 2025: they are not single-event long shots, and they make up 3.05 million of 4.00 million qualifying markets. No result had been seen.
5. **Hourly series (30 October).** Series that reset every hour or fifteen minutes are excluded, as in the benchmark study, from every test. Seen before the change: the validation with and without the filter; no other result.
6. **Disclosure (30 October).** While checking the downloaded data, the share of 1–10 cent contracts that settled Yes after November 2025 (3.7%, against an average price of 3.9 cents) was computed before the tests. This is close to the input of H1 for that period, so H1 cannot be called fully blind there; no choice was changed after it.
7. **Polymarket (30 October).** Sample: order-book markets from March 2023 to January 2026, legacy automated-market-maker trades excluded, each trade counted once. Adaptations: the long shot is whichever outcome costs 1–10 cents; the close is the last trade; clusters are days. No result had been seen.
8. **Maker side of the grid (30 October).** For makers, the three entry times are trades made at least 0 hours, 1 day or 7 days before the close; H2 had been computed.
9. **Exploratory splits (30 October).** H1 by freshness of the entry price and by liquidity, added after H1, H2 and the fund had been computed, and reported as exploratory.

10. **Part B data (30 October).** Every S&P 500 and Nasdaq-100 range contract from November 2025 was downloaded without a volume filter, with prices 18, 12 and 6 hours before the close as robustness entries; no Part B result had been computed.
11. **Part B prices (30 October).** Part B is also reported with prices from trades in the previous 24 hours, and on days when those prices add up to 0.8–1.2, after the pre-registered results had been seen; the pre-registered version remains primary.
12. **Correction (30 October).** An earlier validation run computed fees for single contracts; the plan's 100-contract fee is used throughout, and both versions of the validation are reported in Table 2.
13. **p-values (31 October).** After a review of the first draft, every p-value is a percentile p-value (the share of bootstrap draws on the wrong side of zero) instead of a centred one, and clusters are sorted before resampling, so that p-values and intervals cannot disagree and the same test gives the same p-value in every table. All results had been seen; five Polymarket maker specifications (two with fees, three without) changed from *Supported* to *Inconclusive* as a result, so that all 24 Polymarket maker specifications (12 with fees, 12 without) are now *Inconclusive*.
14. **Review checks (31 October).** Added after all results had been seen, and reported as checks rather than tests: a comparison of the two Kalshi price sources on the same markets, executable bid and ask prices, scheduled rather than recorded closing times, returns weighted by trading volume, the Polymarket fund's returns by liquidity, and the insurance's payouts on the worst market days. Hypothesis labels for H3 and H4 were also brought in line with the stated rules.
15. **H3 label rule (31 October).** The plan's H3 condition (negative skewness and a worst month more than six times the median month) is met almost automatically by a fund that is mostly in cash. After a second review, H3 is labelled *Supported* only if the skewness is also significantly negative (its 95% interval below zero). Polymarket's H3 changed from *Supported* to *Inconclusive* (skewness −0.02, interval −0.50 to 0.69); Kalshi's remains *Contradicted*.
16. **Date (31 October).** The paper is dated 31 October 2026, the date of its last revision; the plan (27 October) and Part B (30 October) keep their own dates, and every change made on 31 October is listed above.

APPENDIX D. DATA PROCESSING

- Kalshi trade data: settled binary markets joined with their trades; for each market, the last trade at or before 1 hour and 0 to 10 days before the close, and the contracts traded in the 24 hours before each point. Trades at 1–20 and 80–99 cents were kept by market, price, taker side and day for the maker tests.
- Kalshi price histories: settled binary markets closing after 23 November 2025; hourly candles for the 12 days before the close; the last traded price at or before each entry time. Prices and volumes from Kalshi's archive and from its recent-market store are labelled differently and both are read.
- Index range contracts: every S&P 500 and Nasdaq-100 daily range contract from 24 November 2025 to 27 October 2026, requested day by day; each day has exactly one winning range.
- Polymarket: order-fill records from the blockchain, matched to resolved markets by token; one record per trade is used for volumes, and the taker's own record identifies the side taken; block numbers are converted to times.
- Check sample (review): 1,500 long-shot markets drawn at random from the H1 sample before November 2025 and 1,500 after; in the comparison of price sources, 43 of the earlier markets returned no hourly price history and 17 had no hourly price before the entry time, leaving 1,440; for each, Kalshi's market record (with its expected expiration) and its hourly price history, with the last price, bid and ask 24 hours and 1 hour before the close.
- All analysis is in Python with DuckDB; the code reproduces every table and figure from the processed files.